

Data Analysis with Python

Lesson 5: Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) - Study Notes

Key Concepts

- **Descriptive Statistics** – mean, median, mode, variance, standard deviation, percentiles, and interquartile range (IQR).
- **Correlation & Association** – Pearson, Spearman, and Kendall coefficients; visualizing relationships with heatmaps and pair plots.
- **Data Visualization** – histograms, boxplots, violin plots, scatter plots, and Seaborn's `pairplot`/`jointplot``.
- **Data Cleaning & Preparation** – handling missing values, outliers, and categorical encoding before analysis.
- **Pandas & Seaborn Integration** – leveraging Pandas' `describe()``, `corr()``, and Seaborn's plotting functions for quick, informative visual summaries.

Summary

Exploratory Data Analysis (EDA) is the first, often most crucial, step in any data science workflow. It involves summarizing the main characteristics of a dataset, usually through quantitative measures and visualizations, to uncover patterns, spot anomalies, and form hypotheses. In Python, Pandas provides the backbone for data manipulation and quick statistical summaries, while Seaborn (built on Matplotlib) offers a high level interface for elegant, informative plots.

The typical EDA process starts with a quick glance at the data's shape, data types, and missingness. Next, we compute descriptive statistics to understand central tendency and dispersion. Correlation matrices reveal linear relationships between numeric variables. Visual tools—histograms for distribution, boxplots for outliers, scatter plots for bivariate relationships—help translate numbers into intuition. Throughout, we iterate: clean data, recompute, and refine visualizations until the dataset's story becomes clear.

Important Points

- **Use `df.describe()``** for a fast snapshot of numeric columns; add `include='all`` for categorical stats.
- **Missing data**: `df.isnull().sum()`` shows counts; consider imputation (`fillna``, `interpolate``) or dropping (`dropna``).
- **Outliers**: Boxplots and IQR (`Q3 - Q1``) help flag extreme values; decide whether to cap, transform, or remove them.
- **Correlation**: `df.corr(method='pearson)`` gives Pearson coefficients; for non linear monotonic relationships, use `spearman``.
- **Heatmap**: `sns.heatmap(corr, annot=True, cmap='coolwarm)`` visualizes correlation matrices clearly.

- **Pair Plot**: `sns.pairplot(df, hue='target')` shows pairwise relationships and class distribution simultaneously.
- **Joint Plot**: `sns.jointplot(x='x', y='y', data=df, kind='reg')` overlays regression line on scatter plot.
- **Seaborn Themes**: `sns.set_style('whitegrid')` or `sns.set_context('talk')` improve readability.
- **Data Types**: Convert categorical columns to `category` dtype to save memory and enable Seaborn's hue mapping.
- **Reproducibility**: Keep a notebook or script that logs every cleaning step; EDA is iterative, so version control helps.

Examples

1. Summary Statistics & Correlation Heatmap

```
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

## Load example dataset

```
df = sns.load_dataset('tips')
```

### 1. Descriptive stats

```
print(df.describe())
```

### 2. Correlation matrix

```
corr = df.corr()
print(corr)
```

### 3. Heatmap

```
plt.figure(figsize=(6,4))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap of Tips Dataset')
plt.show()
```
```

***Explanation*:**

- `df.describe()` shows mean, std, min, max, etc.
- `df.corr()` computes Pearson correlation for numeric columns.
- The heatmap visualizes these relationships, making it easy to spot strong positive or negative associations (e.g., `total_bill` vs. `tip`).

2. Pair Plot with Hue and Missing Value Handling

```
```python
```

## Introduce missing values for demonstration

```
df.loc[df.sample(frac=0.1).index, 'total_bill'] = None
```

## Handle missing values by dropping rows

```
df_clean = df.dropna(subset=['total_bill'])
```

## Pair plot colored by day of the week

```
sns.pairplot(df_clean, hue='day', diag_kind='kde')
plt.suptitle('Pair Plot of Tips Dataset', y=1.02)
plt.show()

```

\*Explanation\*:

- Missing values are removed before plotting to avoid NaNs in Seaborn.
- `pairplot` shows distribution on diagonals (`kde`) and scatter plots for each pair of variables.
- `hue='day'` colors points by day, revealing potential day of week effects.

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## Quick Review

1. What Pandas function gives a quick summary of numeric columns?
2. Which Seaborn function is best for visualizing a correlation matrix?
3. How can you identify outliers using a boxplot?
4. Name two correlation methods available in Pandas and when you might use each.
5. Why is it important to convert categorical columns to the `category` dtype before plotting?

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## Further Reading

- **Data Cleaning**: Handling missing values, outlier treatment, and encoding categorical variables.
- **Advanced Visualizations**: Seaborn's `catplot`, `violinplot`, and `boxenplot` for richer categorical comparisons.
- **Dimensionality Reduction**: PCA, t-SNE, and UMAP for visualizing high dimensional data.
- **Statistical Testing**: Hypothesis tests (t-test, chi-square) to validate patterns seen in EDA.
- **Automated EDA Tools**: `pandas\_profiling`, `sweetviz`, and `dtale` for quick, report-style summaries.

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